

WEEK 7 – CAPITAL BUDGETING (EXAM-READY NOTES)

Source: WEEK 7 Capital Budgeting Notes

1. INTRODUCTION TO CAPITAL BUDGETING

Definition

Capital Budgeting is the process of evaluating and selecting long-term investment projects. It involves deciding how a firm should invest its current funds in long-term assets to generate future benefits.

Key Definition from Notes

"Capital Budgeting is defined as the firm's decision to invest its current funds most efficiently in long-term activities in anticipation of an expected flow of future benefits over a series of years."

Important Points

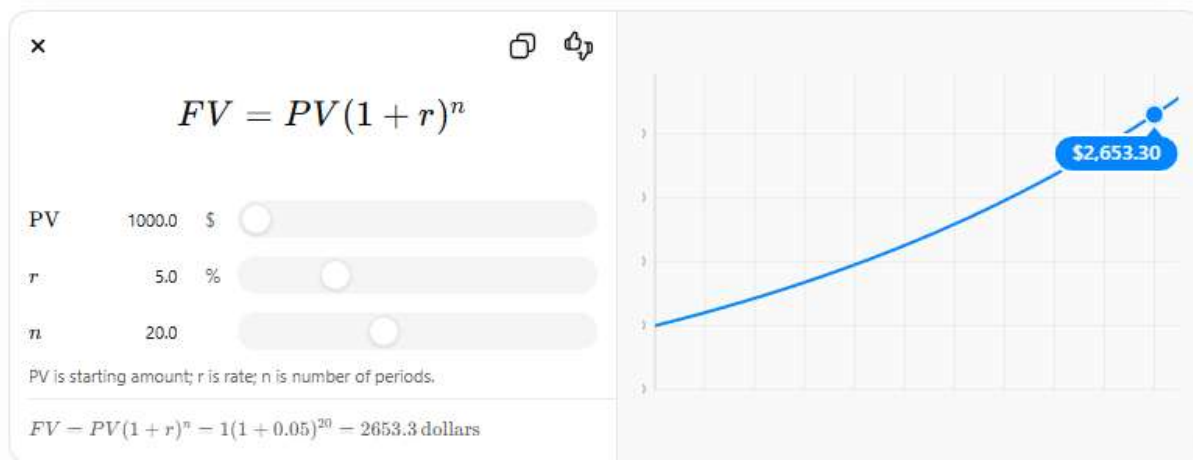
- Concerned with long-term investments.
 - Also called **Project Appraisal**.
 - Deals with:
 - Addition of fixed assets
 - Replacement of assets
 - Modification of assets
 - Expansion projects
 - Decisions affect the company for many years.
-

2. TIME VALUE OF MONEY IN CAPITAL BUDGETING

Capital budgeting heavily relies on the **Time Value of Money (TVM)** concept.

Compounding Factor (Future Value)

Used to calculate future value of present cash.

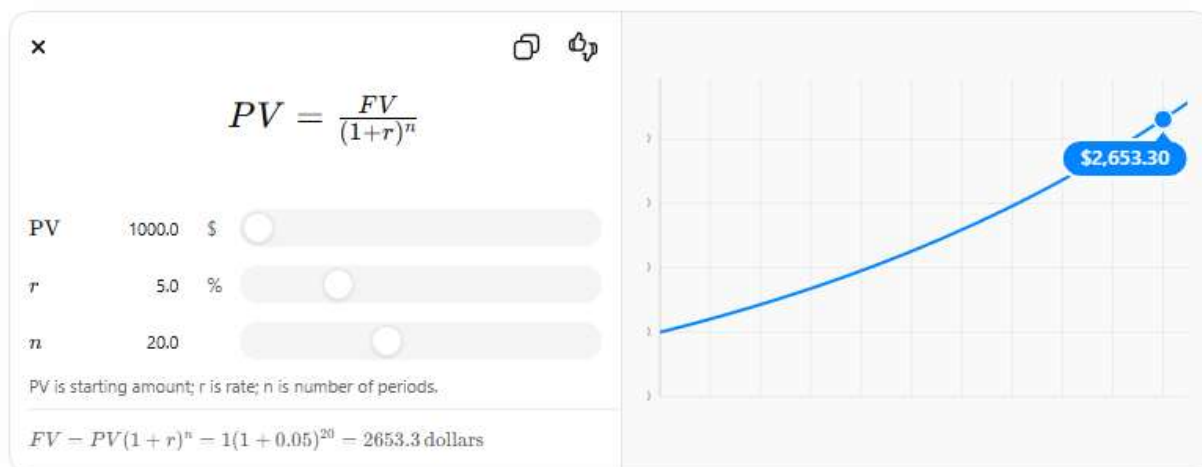


Where:

- FV = Future Value
- PV = Present Value
- r = Interest Rate
- n = Number of Years

Discounting Factor (Present Value)

Used to calculate present value of future cash flows.



3. NEED FOR CAPITAL BUDGETING

According to the notes, capital budgeting decisions are required for:

Purpose	Explanation
Expansion	Increase production capacity
Replacement	Replace obsolete machinery
Diversification	Enter new business/product lines
Buy or Lease	Decide ownership vs leasing
Research & Development	Invest in innovation and technology

A. Expansion

- Increase production facilities.
- Meet future market demand.
- Requires:
 - New plant
 - Machinery
 - Additional working capital

B. Replacement

Reasons:

- Machinery wears out.
- Technology becomes obsolete.
- Productivity decreases.

Purpose:

- Modernization
- Renovation
- Efficiency improvement

C. Diversification

- Entry into new products/businesses.
- Requires significant long-term investment.

D. Research & Development

- Development of new technology.
 - Innovation for future growth.
 - Long-term benefits expected.
-

4. TYPES OF CAPITAL INVESTMENT DECISIONS

The diagram on Page 2 classifies capital investment decisions into two categories.

Based on Firm's Existence

1. Replacement & Modernization Decisions

Replacing old assets.

2. Expansion Decisions

Increasing business capacity.

3. Diversification Decisions

Entering new businesses.

Based on Decision Situation

1. Mutually Exclusive Decisions

Selecting one project among alternatives.

2. Accept-Reject Decisions

Approve or reject individual projects.

3. Contingent Decisions

One project depends on another.

5. CAPITAL BUDGETING TECHNIQUES

The chart on Page 4 divides techniques into two major categories.

Traditional (Non-Discounting)	Discounted Cash Flow Methods
Payback Period	NPV

Traditional (Non-Discounting)	Discounted Cash Flow Methods
ARR	Profitability Index
	IRR
	Discounted Payback Period
	MIRR

6. PAYBACK PERIOD (PBP)

Meaning

Payback Period is:

"The number of years required to recover the original investment from cash inflows."

Formula

For uniform cash flows:

$$\text{Payback Period} = \frac{\text{Initial Investment}}{\text{Annual Cash Inflow}}$$

Decision Rule

Condition	Decision
Shorter PBP	Accept
Longer PBP	Reject

Advantages

- Simple to calculate
- Easy to understand
- Focuses on liquidity

Limitations

- Ignores TVM
 - Ignores cash flows after payback period
-

7. ACCOUNTING RATE OF RETURN (ARR)

Meaning

Measures profitability using accounting profit (PAT).

Formula

$$ARR = \frac{\text{Average Annual PAT}}{\text{Initial Investment}} \times 100$$

Decision Rule

Condition	Decision
ARR > Required Rate	Accept
ARR < Required Rate	Reject

Advantages

- Easy calculation
 - Uses accounting data
-

Limitations

- Ignores TVM
 - Based on accounting profit rather than cash flows
-

8. NET PRESENT VALUE (NPV)

Meaning

NPV is the difference between:

- Present Value of Cash Inflows
 - Present Value of Cash Outflows
-

Formula

$$\text{NPV} = \text{PV of Inflows} - \text{Initial Investment}$$

Decision Rules

NPV Value	Decision
NPV > 0	Accept
NPV = 0	Indifferent
NPV < 0	Reject

Significance

- Considers Time Value of Money.
 - Maximizes shareholder wealth.
 - Most reliable method.
-

9. PROFITABILITY INDEX (PI)

Meaning

Also called:

Benefit-Cost Ratio (BCR)

Formula

$$PI = \frac{\text{PV of Future Cash Inflows}}{\text{Initial Investment}}$$

Decision Rule

PI	Decision
$PI > 1$	Accept
$PI = 1$	Indifferent
$PI < 1$	Reject

10. INTERNAL RATE OF RETURN (IRR)

Meaning

IRR is:

The discount rate at which NPV becomes zero.

Formula Concept

$NPV = 0$

Acceptance Rule

Condition	Decision
$IRR > \text{Cost of Capital (k)}$	Accept
$IRR < \text{Cost of Capital (k)}$	Reject
$IRR = \text{Cost of Capital (k)}$	Neutral

Importance

- Considers TVM
 - Useful for project ranking
 - Popular investment appraisal tool
-

11. CASH FLOW COMPUTATION (VERY IMPORTANT)

The notes provide a standard format for capital budgeting problems.

Step	Calculation
Net Income	Given
Less: Depreciation	NI – Depreciation
Profit Before Tax	PBT
Less: Tax	Tax
Profit After Tax	PAT
Add: Depreciation	Non-cash expense
Cash Flow After Tax	CFAT

Formula

CFAT = PAT + Depreciation

12. DISCOUNT FACTOR

Formula

$$DF = \frac{1}{(1 + r)^n}$$

Example from notes (10% discount rate):

Year	DF
1	0.909
2	0.826
3	0.751
4	0.683
5	0.621

13. NUMERICAL RESULTS FROM CLASS EXAMPLE

Project Investment = ₹20,00,000

Results obtained in the lecture example:

Technique	Result	Decision
Payback Period	3 Years 10.8 Months	Accept
Discounted Payback	4 Years 8.72 Months	Accept
ARR	19.6%	Accept
NPV	₹1,52,900	Accept
PI	1.076	Accept
IRR	12.49%	Accept

EXPECTED EXAM QUESTIONS WITH ANSWERS

Q1. What is Capital Budgeting?

Answer:

Capital budgeting is the process of evaluating long-term investment projects and selecting projects that maximize shareholder wealth by generating future benefits.

Q2. Explain the need for Capital Budgeting.

Answer:

Capital budgeting is required for:

- Expansion
 - Replacement
 - Diversification
 - Buy or lease decisions
 - Research and development
-

Q3. What is Payback Period?

Answer:

Payback period is the number of years required to recover the original investment from project cash inflows.

Q4. Explain NPV and its decision rule.**Answer:**

NPV is the difference between present value of inflows and investment cost.

- $NPV > 0 \rightarrow \text{Accept}$
 - $NPV = 0 \rightarrow \text{Indifferent}$
 - $NPV < 0 \rightarrow \text{Reject}$
-

Q5. Explain IRR.**Answer:**

IRR is the discount rate at which NPV becomes zero. Projects are accepted when IRR exceeds the cost of capital.

 CASE STUDY-BASED ANSWERS (FINANCIAL MANAGEMENT)**Case 1: Expansion Decision**

Situation: A manufacturing company wants to increase production capacity.

Answer:

This is an Expansion Capital Budgeting Decision. Management should evaluate projects using NPV and IRR. Projects with positive NPV and IRR greater than the cost of capital should be accepted.

Case 2: Replacement of Machinery

Situation: Existing machines have become obsolete.

Answer:

This is a Replacement Decision. The company should compare:

- Cost of new machinery
- Additional cash inflows
- Cost savings

NPV and IRR should be used before replacing the machinery.

Case 3: Diversification Project

Situation: A company wants to enter a new product line.

Answer:

Diversification requires large capital investment. The project should be accepted only if:

- $NPV > 0$
 - $PI > 1$
 - $IRR > \text{Cost of Capital}$
-

Case 4: Project Selection

Situation: Project A has $IRR = 14\%$, $\text{Cost of Capital} = 10\%$.

Answer:

Since $IRR > \text{Cost of Capital}$, the project generates returns above the required rate and should be accepted.

Case 5: NPV Analysis

Situation: A project shows $NPV = -₹2,00,000$.

Answer:

Negative NPV indicates that the project fails to recover the required return. Therefore, the project should be rejected.



LAST-MINUTE REVISION FORMULAS

$$FV = PV(1 + r)^n$$

$$PV = \frac{FV}{(1 + r)^n}$$

$$DF = \frac{1}{(1 + r)^n}$$

$$ARR = \frac{\text{Average } PAT}{\text{Initial Investment}} \times 100$$

$$NPV = PV \text{ Inflows} - \text{Investment}$$

$$PI = \frac{PV \text{ Inflows}}{\text{Investment}}$$

$$IRR = \text{Rate at which } NPV = 0$$

$$CFAT = PAT + \text{Depreciation}$$

Exam Priority Ranking

Very Important: NPV, IRR, PI, CFAT Calculation

Important: ARR, Payback Period, Discount Factor

Theory Questions: Need for Capital Budgeting, Types of Investment Decisions, Expansion vs Replacement vs Diversification.

WEEK 9 – CAPITAL STRUCTURE (EXAM-READY SUMMARY)

Source: WEEK 9 Capital Structure Notes

1. INTRODUCTION TO CAPITAL STRUCTURE

Meaning

The term **Capital Structure** refers to the composition or mix of long-term sources of finance used by a firm.

It includes:

- Equity Share Capital
- Preference Share Capital
- Debentures
- Long-Term Loans
- Retained Earnings

Definition

Capital Structure refers to the proportions or combinations of equity share capital, preference share capital, debentures, long-term loans, retained earnings and other long-term sources of funds in the total amount of capital raised by a firm.

2. IMPORTANCE OF CAPITAL STRUCTURE

A sound capital structure helps maximize the firm's value and shareholder wealth.

Importance	Explanation
Increase Firm Value	Improves market price of shares and overall firm value
Efficient Utilization of Funds	Ensures optimal use of available funds
Maximization of Return	Increases Earnings Per Share (EPS) through trading on equity
Minimization of Cost of Capital	Debt is cheaper due to tax-deductible interest
Solvency & Liquidity	Avoids excessive debt burden
Flexibility	Facilitates future expansion and restructuring
Control	Prevents dilution of ownership control

3. TRADING ON EQUITY (VERY IMPORTANT)

Meaning

Trading on Equity means using **fixed-interest-bearing securities (debt)** to increase returns to equity shareholders.

Key Concept

If:

Return on Capital Employed > Interest Rate

then shareholders earn higher returns.

Example

- ROI = 15%
- Interest on Debt = 10%

Since ROI > Interest, debt financing increases EPS and shareholder return.

4. FACTORS DETERMINING CAPITAL STRUCTURE

The notes identify **12 important factors** affecting capital structure decisions.

1. Risk of Cash Insolvency

- Excessive debt increases fixed interest obligations.
- Failure to pay interest may lead to liquidation.

2. Risk in Variation of Earnings

Higher debt causes:

- Higher financial risk
- Greater fluctuation in EPS

3. Cost of Capital

Finance manager should select sources with lower cost.

Objective: Minimize WACC

4. Control

Debt does not provide voting rights.

Therefore, existing owners may prefer debt to avoid dilution of control.

5. Trading on Equity

Use of debt may increase EPS if ROI exceeds interest rate.

6. Government Policies

Affected by:

- SEBI regulations
 - Monetary policy
 - Fiscal policy
 - Lending policies
-

7. Size of Company

Small Firms	Large Firms
Depend on equity	Can issue various securities
Difficult to borrow	Easier access to debt

8. Needs of Investors

Different investors prefer different securities:

- Risk-averse → Debentures
 - Growth-oriented → Equity shares
-

9. Flexibility

Capital structure should permit future expansion.

10. Period of Finance

Requirement	Source
Permanent	Equity
Long-Term	Debentures / Preference Shares

11. Nature of Business

- Stable earnings → More debt
 - Unstable earnings → More equity
-

12. Legal Requirements

Must comply with Companies Act and regulatory provisions.

5. OPTIMAL CAPITAL STRUCTURE

Meaning

Optimal Capital Structure is the best mix of debt and equity that:

- ✓ Minimizes Cost of Capital (WACC)
 - ✓ Maximizes Firm Value
 - ✓ Maximizes Shareholder Wealth
-

Objective

Firm Value ↑

WACC ↓

6. RISK-RETURN TRADE-OFF

The diagram on Page 9 illustrates the relationship between risk and return.

Principle

Higher Risk = Higher Expected Return

Formula

Required Return = Risk Free Return + Risk Premium

Low Risk Investments

Examples:

- Government Bonds
- Treasury Securities

Characteristics:

- Low return
- Low risk

High Risk Investments

Examples:

- Equity Shares
- Corporate Securities

Characteristics:

- High return potential
- Higher uncertainty

7. THEORIES OF CAPITAL STRUCTURE

The notes discuss four major theories:

1. Net Income (NI) Approach
2. Net Operating Income (NOI) Approach
3. Traditional Approach
4. Modigliani-Miller (MM) Approach

8. NET INCOME (NI) APPROACH

Developed By:

Durand

Main Idea

Increasing debt decreases WACC and increases firm value.

Relationship

Debt	WACC	Firm Value
Increase	Decrease	Increase
Decrease	Increase	Decrease

Assumptions

- Only Debt and Equity exist
 - No taxes
 - No flotation cost
 - Perfect capital market
 - Investors are rational
 - Debt is perpetual
-

Conclusion

Higher leverage leads to:

WACC ↓

Firm Value ↑

9. NET OPERATING INCOME (NOI) APPROACH

Developed By:

Durand

Main Idea

Capital structure does not affect firm value.

Key Features

1. WACC remains constant

WACC = Constant

2. Firm Value

Value of Equity = Firm Value - Debt

3. Cost of Equity Increases

As debt increases:

- Financial risk increases
 - Shareholders demand higher return
-

Conclusion

Capital structure is irrelevant.

10. TRADITIONAL APPROACH

Main Idea

There exists an **Optimal Capital Structure**.

Concept

Initially:

- Debt increases
- WACC decreases

After optimal point:

- Additional debt increases risk
 - WACC rises
-

Graphical Interpretation

WACC forms a U-shaped curve:

- Lowest point = Optimal Capital Structure
-

Assumptions

1. Debt cost remains constant initially.
 2. Cost of equity increases gradually.
 3. Beyond optimum point, cost of equity rises sharply.
-

11. MODIGLIANI & MILLER (MM) APPROACH

Developed By:

Franco Modigliani & Merton Miller

Main Idea

Capital Structure Irrelevance Theory

Principle

Firm value depends on:

✓ Operating Profit (EBIT)

NOT on:

✗ Debt-Equity Mix

Key Statement

The value of a leveraged firm equals the value of an unleveraged firm.

Assumptions

- No taxes
- No transaction cost
- No bankruptcy cost
- Perfect information
- Rational investors

- No flotation cost
- Same borrowing cost for investors and firms

Conclusion

Capital structure does not affect:

- Firm value
- Share price
- WACC

Only operating income matters.

COMPARISON OF THEORIES

Theory	WACC	Firm Value
NI Approach	Decreases with debt	Increases
NOI Approach	Constant	Constant
Traditional Approach	Falls then rises	Maximum at optimum point
MM Approach	Constant	Independent of leverage

EXPECTED EXAM QUESTIONS WITH ANSWERS

Q1. What is Capital Structure?

Answer:

Capital Structure refers to the mix of long-term sources of finance such as equity shares, preference shares, debentures, long-term loans, and retained earnings used by a company.

Q2. Explain Trading on Equity.

Answer:

Trading on Equity is the use of fixed-interest-bearing debt to increase returns to equity shareholders. It is beneficial when ROI exceeds the interest rate on debt.

Q3. What is Optimal Capital Structure?

Answer:

Optimal Capital Structure is the ideal combination of debt and equity that minimizes WACC and maximizes firm value.

Q4. Explain NI Approach.**Answer:**

The Net Income Approach states that increasing debt decreases WACC and increases the value of the firm because debt is a cheaper source of finance.

Q5. Explain MM Theory.**Answer:**

MM Theory states that firm value is independent of capital structure. The value of a firm depends only on its operating income and business risk.

 CASE STUDY–BASED ANSWERS**Case 1: Company Wants to Increase EPS****Situation:**

A company has stable earnings and wants to increase shareholder returns.

Answer:

The company can use Trading on Equity by introducing debt into its capital structure. If ROI exceeds the interest rate, EPS will increase and shareholder wealth will improve.

Case 2: Excessive Debt Financing**Situation:**

A company has borrowed heavily and is struggling to pay interest.

Answer:

The company faces Risk of Cash Insolvency. Excessive leverage increases fixed obligations and may affect solvency. Capital structure should be rebalanced by increasing equity.

Case 3: Expansion Project**Situation:**

A large company plans a major expansion.

Answer:

The firm should adopt an Optimal Capital Structure by balancing debt and equity. The objective should be minimizing WACC while maintaining financial flexibility.

Case 4: Comparing NI and MM Theory**Situation:**

Management wants to know whether debt always increases firm value.

Answer:

According to NI Approach, increasing debt increases firm value. However, MM Theory argues that firm value is independent of debt-equity mix. In practice, firms seek an optimal balance between the two.

 LAST-MINUTE REVISION SHEET**Important Formulas**

$$\text{Required Return} = \text{Risk Free Return} + \text{Risk Premium}$$

$$\text{Value of Equity} = \text{Firm Value} - \text{Debt}$$

$$WACC = \sum W_i K_i$$

Remember

- **Trading on Equity** = Increase debt to increase EPS
- **Optimal Capital Structure** = Minimum WACC + Maximum Firm Value
- **NI Theory** = Debt $\uparrow$ $\rightarrow$ WACC $\downarrow$ $\rightarrow$ Firm Value $\uparrow$
- **NOI Theory** = Capital Structure Irrelevant
- **Traditional Theory** = Optimal Debt-Equity Mix Exists
- **MM Theory** = Firm Value Depends on EBIT, not Financing Mix.

WEEK 11 – DIVIDEND DECISIONS (EXAM-READY SUMMARY)

Source: WEEK 11 Dividend Decisions Notes

1. CONCEPT OF DIVIDEND

A portion of the **net earnings (profits)** of a company distributed to its shareholders is called a **Dividend**.

Definition

According to the **Institute of Chartered Accountants of India (ICAI)**:

"Dividend is a distribution to shareholders out of profits or reserves available for this purpose."

Key Point

Dividend represents the shareholder's share in the profits earned by the company.

2. DIVIDEND DECISION

Meaning

Dividend Decision refers to determining:

- How much profit should be distributed to shareholders.
- How much profit should be retained for future growth and expansion.

Objective

Maintain a balance between:

- Shareholder expectations
 - Business growth requirements
-

3. NATURE OF DIVIDEND

The dividend payment depends upon the **Dividend Policy** adopted by the company.

A. Constant Dividend Per Share (DPS)

Meaning

The company pays a fixed dividend per share every year irrespective of fluctuations in earnings.

Characteristics

- Stable dividend policy.
- Dividend remains constant even during low-profit years.
- Creates investor confidence.

Example

Year	EPS	DPS
1	₹20	₹5
2	₹15	₹5
3	₹10	₹5

B. Constant Pay-out Ratio

Meaning

The company distributes a fixed percentage of profits as dividends.

Formula

$$\text{Dividend Payout Ratio} = \frac{\text{Dividend}}{\text{Net Earnings}} \times 100$$

Characteristics

- Dividend changes with profits.
- Payout ratio remains constant.

Example

If payout ratio = 40%

Profit	Dividend
₹1,00,000	₹40,000
₹2,00,000	₹80,000

C. Stable Cash Dividend Plus Extra Dividend

Meaning

The company pays:

- Regular fixed dividend
- Additional dividend during high-profit years

Features

- Shareholders receive stable income.
 - Bonus dividend paid during exceptionally profitable years.
 - Extra dividend is discontinued if profits decline.
-

4. TYPES OF DIVIDEND

The notes classify dividends on three different bases.

A. BASED ON SOURCE OF PAYMENT

1. Dividend Out of Current Profits

Meaning

Dividend is paid from profits earned during the current financial year.

Important Legal Provision

As per **Section 123 of the Companies Act, 2013**:

Dividend may be declared out of current profits after providing for depreciation.

2. Dividend Out of Retained Earnings

Meaning

Dividend is paid from accumulated profits of previous years.

Features

- Uses undistributed profits.
 - Helps maintain stable dividends during low-profit years.
-

B. BASED ON REGULARITY OF PAYMENT

1. Interim Dividend

Meaning

Dividend declared and paid before the finalization of annual accounts.

Characteristics

- Paid during the financial year.
 - Declared by the Board of Directors.
-

2. Final Dividend

Meaning

Dividend declared after the preparation of annual accounts.

Characteristics

- Announced at Annual General Meeting (AGM).
 - Based on final profits earned during the year.
-

C. BASED ON FORM OF PAYMENT

1. Cash Dividend

Meaning

Dividend paid in cash to shareholders.

Features

- Most common form.
- Causes cash outflow.
- Reduces company liquidity.

Requirement

Company must have sufficient cash balance.

2. Bond Dividend

Meaning

Dividend paid through bonds instead of cash.

Purpose

- Postpones immediate cash outflow.
- Useful when company faces cash shortage.

Note

Not popular in India.

3. Share Dividend (Bonus Shares)**Meaning**

Dividend paid in the form of additional shares.

Features

- No cash outflow.
- Increases number of shares held.

Also Called

- Stock Dividend
 - Bonus Shares
-

4. Property Dividend**Meaning**

Dividend paid through assets other than cash.

Examples

- Land
- Equipment
- Investments

Features

- Rarely used.
 - Not popular in India.
-

5. Scrip Dividend**Meaning**

Dividend paid through a promissory note.

Features

- Company promises future payment.
 - Cash payment deferred.
-

5. COMPARISON OF DIVIDEND POLICIES

Basis	Constant DPS	Constant Payout Ratio	Stable + Extra Dividend
Stability	High	Low	Moderate
Investor Confidence	High	Moderate	High
Dividend Fluctuation	No	Yes	Limited
Growth Flexibility	Moderate	High	High

6. ADVANTAGES OF DIVIDEND POLICY

For Shareholders

- Regular income
- Investor confidence
- Wealth creation

For Company

- Better market reputation
 - Attracts investors
 - Maintains goodwill
-

7. EXPECTED EXAM QUESTIONS WITH ANSWERS

Q1. What is Dividend?

Answer:

Dividend is the portion of company profits distributed among shareholders from current profits or accumulated reserves.

Q2. Explain Constant Dividend Per Share Policy.

Answer:

Under this policy, a company pays a fixed dividend per share every year irrespective of fluctuations in earnings. It provides stability and enhances investor confidence.

Q3. What is Dividend Payout Ratio?**Answer:**

Dividend Payout Ratio represents the percentage of profits distributed as dividends.

$$\text{Dividend Payout Ratio} = \frac{\text{Dividend}}{\text{Net Earnings}} \times 100$$

Q4. Differentiate Interim Dividend and Final Dividend.

Interim Dividend	Final Dividend
Paid during year	Paid after year-end
Declared by Board	Approved at AGM
Based on interim profits	Based on final profits

Q5. Explain Share Dividend.**Answer:**

Share dividend refers to payment of dividend in the form of bonus shares instead of cash. It helps conserve cash while rewarding shareholders.

 CASE STUDY-BASED ANSWERS**Case 1: Company Facing Cash Shortage****Situation**

A company has earned profits but lacks sufficient cash to distribute dividends.

Answer

The company can issue:

- Share Dividend (Bonus Shares)

- Scrip Dividend
- Bond Dividend

This avoids immediate cash outflow while rewarding shareholders.

Case 2: Stable Earnings Company

Situation

A mature company earns consistent profits every year.

Answer

The company should adopt a **Constant Dividend Per Share Policy** because:

- Investors prefer stability.
 - Shareholder confidence increases.
 - Market value of shares improves.
-

Case 3: High Growth Company

Situation

A technology company wants to retain most of its profits for expansion.

Answer

The company should follow a **low dividend payout policy** and retain earnings for future growth and investment opportunities.

Case 4: Exceptional Profit Year

Situation

A company earned unusually high profits during the year.

Answer

The company may follow **Stable Cash Dividend Plus Extra Dividend Policy**, rewarding shareholders without committing to permanently higher dividends.

NUMERICAL PROBLEMS

Problem 1 – Dividend Payout Ratio

Net Profit = ₹5,00,000

Dividend Paid = ₹2,00,000

Dividend Payout Ratio = $(2,00,000 / 5,00,000) \times 100$

Answer = 40%

Problem 2 – Dividend Amount

Net Profit = ₹8,00,000

Dividend Payout Ratio = 25%

Dividend = 8,00,000 * 25%

Answer = ₹2,00,000

Problem 3 – Retained Earnings

Profit = ₹10,00,000

Dividend = ₹4,00,000

Retained Earnings = 10,00,000 - 4,00,000

Answer = ₹6,00,000

LAST-MINUTE REVISION SHEET

Key Definitions

- **Dividend** = Share of profit distributed to shareholders.
- **Interim Dividend** = Paid before year-end.
- **Final Dividend** = Paid after annual accounts.
- **Share Dividend** = Bonus shares issued instead of cash.
- **Scrip Dividend** = Promissory note for future payment.

Important Formula

$$\text{Dividend Payout Ratio} = \frac{\text{Dividend}}{\text{Net Earnings}} \times 100$$

Most Important Exam Topics

- Dividend Policies
- Types of Dividend
- Interim vs Final Dividend
- Dividend Payout Ratio
- Share Dividend vs Cash Dividend
- Case Studies on Dividend Decisions

5-Mark Questions

- Meaning of Dividend
- Types of Dividend
- Dividend Policies

10-Mark Questions

- Explain Dividend Decision and Dividend Policies.
- Discuss Types of Dividend with examples.
- Explain Dividend Payout Ratio and its importance.

WEEK 12 – WORKING CAPITAL MANAGEMENT (EXAM-READY SUMMARY)

Source: WEEK 12 Working Capital Management Notes

1. WORKING CAPITAL (WC)

Meaning

Working Capital refers to the capital used in the day-to-day operations of a business.

It represents the difference between **Current Assets (CA)** and **Current Liabilities (CL)**.

Formula

$$WC = CA - CL$$

Where:

- CA = Current Assets
- CL = Current Liabilities

Working Capital is also known as:

- Net Current Assets
 - Operating Assets
-

2. WORKING CAPITAL MANAGEMENT (WCM)

Meaning

Working Capital Management refers to the management of current assets and current liabilities to ensure efficient business operations and adequate liquidity.

Objective

- Maintain liquidity
- Ensure profitability
- Avoid cash shortages
- Optimize use of funds

Working Capital Management is a strategy designed to monitor and utilize current assets and current liabilities efficiently.

3. NEED FOR WORKING CAPITAL MANAGEMENT

The notes identify three major areas of working capital management:

1. Inventory Management

Managing stock levels efficiently.

2. Receivables Management

Managing credit sales and collections.

3. Cash Management

Maintaining adequate liquidity.

4. FACTORS AFFECTING WORKING CAPITAL

The following factors influence working capital requirements:

Factor	Impact
Nature of Business	Trading firms need less WC than manufacturing firms
Production Policy	Seasonal production increases WC needs
Credit Policy	Liberal credit policy increases WC
Inventory Policy	Higher inventory increases WC
Market Conditions	Demand fluctuations affect WC
Business Cycle	Boom and recession affect WC requirements
Taxation Policy	Higher taxes reduce available funds
Dividend Policy	Dividend payments affect liquidity
Operating Efficiency	Efficient firms need less WC
Raw Material Availability	Scarcity increases inventory holding

5. BUSINESS CYCLE AND WORKING CAPITAL

The diagram on Page 3 shows the business cycle phases:

Stages

1. Boom
2. Recovery

3. Recession
4. Slump

Impact on Working Capital

Phase	WC Requirement
Boom	High
Recovery	Increasing
Recession	Reduced
Slump	Lowest

6. HOW MUCH WORKING CAPITAL IS NEEDED?

The amount of working capital depends upon:

- Size of the firm
- Nature of activities
- Availability of credit
- Attitude toward profit
- Attitude toward risk
- Other operational factors

7. IMPORTANCE OF ADEQUATE / OPTIMUM WORKING CAPITAL

Adequate working capital is essential for efficient operations.

Benefits

1. Smooth running of business
 2. Better profitability with manageable risk
 3. Growth and expansion opportunities
 4. Timely payments
 5. Increased goodwill
 6. Better trade relationships
-

8. TWO MAJOR PROCESSES IN WORKING CAPITAL MANAGEMENT

The diagram on Page 3 highlights two important processes:

1. Forecasting Requirement of Funds

Estimating future working capital needs.

2. Arrangement of Funds

Mobilizing funds from suitable sources.

9. WORKING CAPITAL CYCLE (OPERATING CYCLE)

Meaning

The Working Capital Cycle measures the time taken to convert cash invested in operations back into cash.

The cycle moves through:

Raw Material → Work in Progress → Finished Goods → Debtors → Cash

Formula

Operating\ Cycle = R + W + F + D - C

Where:

Symbol Meaning

R	Raw Material Holding Period
W	Work-in-Progress Period
F	Finished Goods Holding Period
D	Debtors Collection Period
C	Creditors Payment Period

Example from Notes

Given:

- Raw Material = 60 days
- Production = 15 days

- Finished Goods = 30 days
- Debtors = 30 days
- Creditors = 15 days

$$60 + 15 + 30 + 30 - 15$$

$$= 120 \text{ Days}$$

Working Capital Cycle = 120 Days

10. COMPONENTS OF OPERATING CYCLE

A. Raw Material Storage Period

$$\frac{\text{Average Stock of Raw Material}}{\text{Average Cost of Raw Material Consumed per Day}}$$

B. Work-in-Progress Holding Period

$$\frac{\text{Average WIP Inventory}}{\text{Average Cost of Production per Day}}$$

C. Finished Goods Storage Period

$$\frac{\text{Average Stock of Finished Goods}}{\text{Average Cost of Goods Sold per Day}}$$

D. Debtors Collection Period

$$\frac{\text{Average Debtors}}{\text{Average Credit Sales per Day}}$$

E. Credit Period Allowed by Suppliers

$$\frac{\text{Average Creditors}}{\text{Average Credit Purchases per Day}}$$

11. WORKING CAPITAL POLICIES / APPROACHES

The notes discuss two major approaches.

A. Conservative Approach

Characteristics

- Uses more long-term financing.
- Low risk.
- High liquidity.
- Lower profitability.

Advantages

- ✓ Financial stability
- ✓ Lower insolvency risk

Disadvantage

- ✗ Lower returns
-

B. Aggressive Approach

Characteristics

- Relies more on short-term financing.
- Higher profitability.
- Higher risk.

Advantages

- ✓ Higher returns

Disadvantages

- ✗ Greater liquidity risk
 - ✗ Higher refinancing risk
-

Comparison

Basis	Conservative	Aggressive
Risk	Low	High
Profitability	Low	High
Liquidity	High	Low
Financing Source	Long-term	Short-term

12. FINANCING OF WORKING CAPITAL

Working Capital can be financed through:

A. Long-Term Sources

1. Equity Share Capital
 2. Preference Share Capital
 3. Debentures
 4. Bonds
 5. Bank & Financial Institution Loans
 6. Retained Earnings
 7. Venture Capital
-

B. Short-Term Sources

1. Bank Credit
 2. Trade Credit
 3. Advances from Customers
 4. Bank Advances
 5. Loans
 6. Overdrafts
 7. Bills Discounting
 8. Commercial Paper
 9. Bank Deposits
-

13. REGULATION OF BANK FINANCE

Banks are major providers of working capital finance.

The RBI appointed committees to improve working capital financing practices.

14. DAHEJA COMMITTEE

Objective

To improve bank lending practices.

Findings

- No relationship between optimum WC requirement and bank loans.
- Excessive dependence on multiple financing.

Recommendation

Move toward:

Project-based and need-based lending.

15. TANDON COMMITTEE

Constituted by RBI in 1974 under **P. L. Tandon**.

Objectives

- Determine working capital requirements.
 - Assess end-use of credit.
 - Improve credit discipline.
 - Develop lending norms.
-

Key Recommendations

1. Banks should finance genuine working capital needs.

2. Determine Working Capital Gap

Working Capital Gap = Current Assets - Current Liabilities

3. Borrowers should contribute a portion of working capital.

4. Establish norms for bank lending.

EXPECTED EXAM QUESTIONS WITH ANSWERS

Q1. What is Working Capital?

Answer:

Working Capital is the difference between Current Assets and Current Liabilities.

$$WC = CA - CL$$

Q2. Define Working Capital Management.

Answer:

Working Capital Management is the process of managing current assets and current liabilities efficiently to maintain liquidity and profitability.

Q3. Explain the Operating Cycle.

Answer:

Operating Cycle is the time required to convert raw materials into cash through production, sales, and collection from debtors.

$$\text{Operating Cycle} = R + W + F + D - C$$

Q4. Explain Conservative and Aggressive Approaches.

Answer:

- Conservative Approach uses more long-term financing and involves low risk.
 - Aggressive Approach uses more short-term financing and involves higher risk but higher returns.
-

Q5. State the recommendations of the Tandon Committee.

Answer:

- Finance genuine working capital needs.
- Determine working capital gap.

- Establish lending norms.
 - Improve credit discipline.
-

CASE STUDY–BASED ANSWERS

Case 1: Manufacturing Company Facing Liquidity Problems

Situation

The company cannot meet short-term obligations.

Answer

The company should improve working capital management by:

- Reducing inventory levels.
 - Speeding up receivable collections.
 - Increasing short-term financing.
 - Improving cash management.
-

Case 2: Fast Growing Business

Situation

Sales are increasing rapidly.

Answer

The company needs additional working capital to finance:

- Inventory
- Receivables
- Operating expenses

A Conservative Approach may be adopted to ensure liquidity.

Case 3: Long Working Capital Cycle

Situation

Operating cycle exceeds industry average.

Answer

The firm should:

- Reduce inventory holding period.
 - Improve production efficiency.
 - Accelerate debtor collections.
 - Negotiate longer credit from suppliers.
-

Case 4: Choosing Financing Approach

Situation

A company wants higher profitability despite higher risk.

Answer

The Aggressive Working Capital Policy is suitable because it relies more on short-term financing and offers higher returns.

NUMERICAL PROBLEMS

Problem 1 – Working Capital

Current Assets = ₹8,00,000

Current Liabilities = ₹3,00,000

WC = 8,00,000 - 3,00,000

Answer = ₹5,00,000

Problem 2 – Operating Cycle

Raw Material = 50 days

WIP = 20 days

Finished Goods = 30 days

Debtors = 40 days

Creditors = 25 days

OC = 50 + 20 + 30 + 40 - 25

Answer = 115 Days

Problem 3 – Debtors Collection Period

Average Debtors = ₹6,00,000

Annual Credit Sales = ₹36,00,000

Credit Sales per Day = ₹10,000

$$\text{Debtors Collection Period} = \frac{6,00,000}{10,000}$$

Answer = 60 Days



LAST-MINUTE REVISION SHEET

Key Formulas

$$WC = CA - CL$$

$$\text{Operating Cycle} = R + W + F + D - C$$

$$\text{Working Capital Gap} = CA - CL$$

Most Important Topics

- Working Capital Cycle
- Operating Cycle Formula
- Conservative vs Aggressive Approach
- Sources of Working Capital Finance
- Tandon Committee
- Daheja Committee

10-Mark Questions

1. Explain Working Capital Management and Operating Cycle.
2. Discuss Working Capital Policies.
3. Explain Financing of Working Capital.
4. Discuss the recommendations of the Tandon Committee.